MTH 200 Chapter 4: Applications of Derivatives

4.10 Antiderivatives

Motivation: Suppose a particle moving in straight line has velocity function $v(t)=t^2+2t$. Suppose $s(t)$ is the position function describing the motion of the particle. You are told that $s(0)=0$. How might we find a formula for $s(t)$?

We know that the velocity function satisfies $v(t)=s'(t)$. So, we need to *undo* differentiation to obtain $s(t)$. In other words the formula for $s(t)$ needs to have derivative equal to $v(t)=t^2+2t$.

After some thought one should notice that $\frac{t^3}{3}+t^2$ has first derivative equal to $v(t)=t^2+2t$.

So, we suspect that $s(t)=\frac{t^3}{3}+t^2$.

We note that this formula works perfectly since with this formula we also have that $s(0)=0$.

Question: Is $s(t)=\frac{t^3}{3}+t^2$ *the only formula which has a derivative of* $v(t)=t^2+2t$*?*

We will now formalize the notions discussed in the motivation above.

Definition: A function F is called the **antiderivative** of f on an interval I if $F'(x)=f(x)$ for all x in I.

Theorem: If F is an antiderivative of f on an interval I, then the most general antiderivative of f on I is

$$F(x)+C$$

where C is an arbitrary constant.

Example 1: Find the most general antiderivative of each of the following functions.

a) $f(x)=\sin(x)$

$$F(x)=-\cos(x)+C$$

b) $f(x)=x^n$ (where n is a real number)

$$F(x)=\frac{x^{n+1}}{n+1}+C$$

c) $f(x)=\frac{1}{x}$

$$F(x)=\ln|x|+C$$

d) $f(x)=e^x$

$$F(x)=e^x+C$$

Note: It is very important, especially when first starting to work with antiderivatives, to check your answer by ensuring that your final answer differentiates to the function with which you started.

<u>Example 2:</u> Find the most general antiderivative of each of the following functions.

a) $f(x)=\sec^2(x)$

$$F(x)=\tan(x)+C$$

b) $f(x)=2^x\ln(2)$

$$F(x)=2^x+C$$

c) $f(x)=\cos(x)$

$$F(x)=\sin(x)+C$$

d) $f(x)=\frac{1}{1+x^2}$

$$F(x)=\tan^{-1}(x)+C$$

e) $f(x)=\csc(x)\cot(x)$

$$F(x)=-\csc(x)+C$$

Theorem: If F and G are antiderivatives of f and g respectively, then

- $c \cdot F(x)$ is an antiderivative of $c \cdot f(x)$
- $F(x)+G(x)$ is an antiderivative of $f(x)+g(x)$.

Note: This theorem just highlights that we can use the Constant Multiple Rule and the Sum & Differences Rules in a "backwards" direction.

Example 3: Find the most general antiderivative of $f(x)=6x^2-2x+1$.

Using the theorem above we see that we can find an antiderivative for each term in function. Therefore we obtain:

$$F(x)=2x^3-x^2+x+C$$

Example 4: Find the most general antiderivative of $g'(x)=4\cos(x)+\dfrac{2x^5-\sqrt{x}}{x}$.

Before we begin find the antiderivative we will use some algebra to rewrite the given derivative.
Note that:

$$g'(x)=4\cos(x)+\frac{2x^5-\sqrt{x}}{x}=4\cos(x)+\frac{2x^5}{x}-\frac{\sqrt{x}}{x}=4\cos(x)+2x^4-x^{-1/2}$$

Therefore we compute that:

$$g(x)=4\sin(x)+\frac{2}{5}x^5-2x^{1/2}+C$$

Example 5: Find *the* function f if $f'(x)=3x^5-2x^3+6x+1$ and $f(1)=3$.

We first note that:

$$f(x)=\frac{1}{2}x^6-\frac{1}{2}x^4+3x^2+x+C$$

Using the given information that $f(1)=3$ we can try to find the exact value of the mystery constant C.

$$3=\frac{1}{2}(1)^6-\frac{1}{2}(1)^4+3(1)^2+(1)+C \quad \Leftrightarrow \quad 3=\frac{1}{2}-\frac{1}{2}+3+1+C \quad \Leftrightarrow \quad C=-1$$

Therefore we can say that:

$$f(x)=\frac{1}{2}x^6-\frac{1}{2}x^4+3x^2+x-1$$

Note: Problems like Ex#5 (and the motivating example) are called ***initial value problems*** since we are told a value of the original function.

Definition: An equation which includes a function and its derivative(s) is called a **differential equation.**

Ex: $y''+3y'+y=x$ & $\dfrac{dy}{dx}=4y$ & $\dfrac{dy}{dx}=6x+2$

Engineering majors and some other majors require a course in differential equations. They will need to analyze a differential equation by finding a formula, a graph, or numerical estimates for a function *y* which satisfies the differential equation.

Example 6: Find *f* if $f'(x)=e^x+20(1+x^2)^{-1}$ and $f(0)=-2$.

We first note that:

$$f'(x)=e^x+20(1+x^2)^{-1}=e^x+\frac{20}{1+x^2}$$

Now we compute:

$$f(x)=e^x+20\tan^{-1}(x)+C$$

Using the given information that $f(0)=-2$ we can try to find the exact value of the mystery constant *C*.

$$-2=e^0+20\tan^{-1}(0)+C \quad\Leftrightarrow\quad -2=1+0+C \quad\Leftrightarrow\quad C=-3$$

Therefore we can say that:

$$f(x)=e^x+20\tan^{-1}(x)-3$$

Example 7: A particle moves in a straight line and has acceleration given by $a(t)=3t^3+3\cos(t)$. The velocity of the particle at time 0 is 7 *m/s* and its position at time 0 is 10 *meters* to the right of the origin. Find the position function $s(t)$ for the particle.

We know that $a(t)=3t^3+3\cos(t)$. We can find the velocity equation by finding an antiderivative.

$v(t)=\dfrac{3}{4}t^4+3\sin(t)+C$ We are told that $v(0)=7$ so $7=\dfrac{3}{4}(0)^4+2\sin(0)+C \quad\Leftrightarrow\quad C=7$. So,

$$v(t)=\frac{3}{4}t^4+3\sin(t)+7$$

Now we can find the position equation by finding another antiderivative.

$$s(t)=\frac{3}{20}t^5-3\cos(t)+7t+D$$

We are told that $s(0)=10$ so $10=\dfrac{3}{20}(0)^4-3\cos(0)+7(0)+D \quad\Leftrightarrow\quad 10=-3+D \quad\Leftrightarrow\quad D=13$. So,

$$s(t)=\frac{3}{20}t^5-3\cos(t)+7t+13$$

Note: This completes section 4.10. At this point it is easy to get the wrong impression about antiderivatives. In general, finding the antiderivative of a function can be very difficult or sometimes even impossible! On the other hand, differentiation can always be applied in a standard way to the functions of calculus.

Note: To get an idea of what I mean in the note above, I would dare you to try to find antiderivatives of the functions: $f(x)=\ln(x)$, $g(x)=\sin^{-1}(x)$, *and* $h(x)=e^{-x^2}$.